Earthquake:
Understanding Earthquakes and School Preparedness



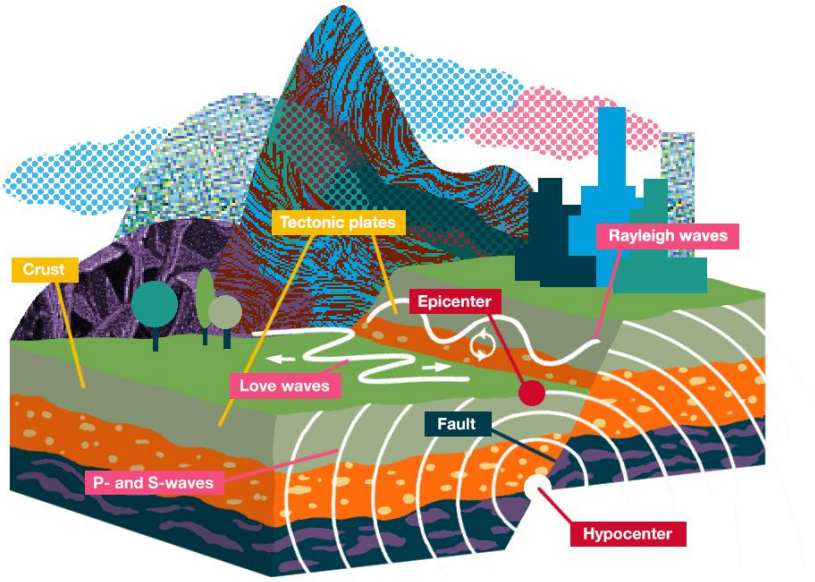
Objectives:

After completing this chapter, you will be able to:

- ✧ Define what an earthquake is.
- ✧ Identify possible effects of an earthquake.
- ✧ Explain why earthquake preparedness is important.
- ✧ Recognize common hazards in the school environment.
- ✧ Describe what to feel, see, and hear during an earthquake.
- ✧ Learn basic safety responses during an earthquake

What is an earthquake?

An earthquake is a weak to violent shaking of the ground produced by the sudden movement of rock materials below the earth's surface.



What are the possible effects of an earthquake?



Strong ground shaking can cause injuries to people due to broken glasses and fallen objects. Buildings may be damaged. Improperly constructed buildings may collapse, resulting in serious injury or death.

Why do we need to prepare for an earthquake?
Earthquakes are unpredictable, so it's important to teach students how to respond to stay safe. Since children spend much of their time at school, teachers play a key role in ensuring their safety during such emergencies.

Possible hazards during an earthquake include:

- ✧ Large furniture that could topple (e.g., bookcases, cabinets
- ✧ Hanging items like flowerpots and ceiling fans
- ✧ Glass windows and walls that may shatter
- ✧ Heavy picture frames and mirrors
- ✧ Unsecured cabinet latches
- ✧ Breakable items stored on high or open shelves
- ✧ Obstructions near exits and hallways (e.g., plant boxes, chairs, trash cans)
- ✧ Elevated water tanks and electrical posts

Discuss with students what to expect during an earthquake, such as shaking, falling objects, and potential damage, to help them stay alert and prepared.

WHAT TO TEACH?

You will feel ...

During an earthquake, the shaking can start out gently or feel very strong. It may begin as a slight movement but can quickly become intense enough to knock a person off their feet.

You might see the following during an earthquake:

- ✧ Hanging objects like lamps and wall frames swinging violently or falling
- ✧ Bookshelves, cabinets, tables, and chairs shifting, moving, or toppling over
- ✧ Window panes rattling and possibly breaking

You might hear the following during an earthquake:

- ✧ Low or loud rumbling sounds
- ✧ Shaking and rattling noises
- ✧ Cracking or creaking of wood
- ✧ Breaking glass and other falling objects

VIDEO LINKS:

What is an Earthquake?
https://youtu.be/hkvkcN9rVHs?si=UgTMbDoUPq690uoN

PlateTectonics
https://youtu.be/ryrXAGY1dmE?si=ob-wEmMPG-0LuhKl

Hypocenter and Epicenter
https://youtu.be/IJPgUhX3VSE?si=taCbFOk8p4p_qyXM

Tectonic Plates and Earthquakes
https://youtu.be/FbbwaAIMvvs?si=29IT2aFHD1-JKMTw

Volcanoes, Earthquakes, and Plate Boundaries
https://youtu.be/5wC4rF7YPAM?si=j6c_RurGAl4qY9iO

Reference:

- Philippine Institute of Volcanology and Seismology